Xin Shu

PhD Candidate in Human–Computer Interaction, Open Lab, Newcastle University
x.shu6@newcastle.ac.uk +44 7536 205983 xinshu.info | Google Scholar

Research Profile

HCI researcher studying how people learn with AI and interactive tools. I work with teachers and learners to design, build and evaluate educational systems, using qualitative and quantitative methods. My work spans school education, AI tutoring, music learning and public AI literacy.

Education

Newcastle University, UK — PhD in HCI, in progress *Oct 2023 – Present*
University of Edinburgh, UK — MSc Design Informatics, Merit *Sep 2022 – Sep 2023*
Dalarna University, Sweden — BSc Information Systems, Distinction *Sep 2020 – Jun 2021*
Jiangxi University of Finance and Economics, China — BEng Software Engineering *Sep 2017 – Jun 2021*

Selected Research Experience

FreireVoice / Bridging Contexts — Research Workstream Lead *CHI 2027, under review*
UK–Brazil inclusive AI education collaboration

- Led study design, fieldwork, reflexive thematic analysis and writing for *After the Answer*: 104 students, three public schools and 10 activity-based focus groups, examining how GenAI-supported task completion can leave understanding unresolved.
- Compared focus-group accounts, 10 resource-barrier maps and 96 help-source worksheets to examine learning constraints and access to support.
- Worked with Brazilian teachers on educational game design informed by students' learning conditions and available resources.
- Co-developed guidance for AI use in learning activities, linking permitted assistance to learning objectives and assessment criteria, and asking students to explain and verify AI-supported work.

SocraticRAG — Teacher-Informed AI Tutoring and Evaluation *Jun 2026 – Present*
New York University Abu Dhabi collaboration

- Co-designed an AI tutor with teachers in the UAE; implemented retrieval-augmented generation (RAG) components to ground guidance in course materials.
- Developed a benchmark separating answer leakage, grounding in course content and pedagogical alignment, examining whether guiding questions respond to students' understanding and misconceptions.
- Evaluated three LLMs on 658 synthetic student utterances and 1,974 tutor responses, using rubric-based scoring, Friedman and paired Wilcoxon tests, and human spot-checks to examine response quality and evaluation limitations.

FretFlow / FretMate — Guitar Learning Tools *CHI 2026; IUI 2025*

- Led design and software implementation of FretFlow (haptic, visual and audio guidance) and FretMate (conversational AI support) for guitar learning.
- Co-designed FretFlow with music teachers in the UK, China and Turkey; development included a formative study with five instructors and two rounds of haptic-design refinement.
- Evaluated learning through a 20-participant crossover study (FretFlow) and a 16-learner controlled study (FretMate), each lasting two weeks, combining performance measures, questionnaires and interviews.

ShadowAI — AI Literacy in a Museum *C&C 2026*

- Worked with the International Centre for Life, Newcastle, to design and implement shadow capture and AI-generated image/story displays for a public exhibit.
- Connected hands-on making with comparisons of artefacts and AI interpretations, inviting children and families to discuss uncertainty, errors and the limits of AI.

VR Cultural Heritage Learning *IJHCI 2026*

- Contributed to system design and a 40-participant mixed-methods study comparing active creation with observation, using pre/post knowledge tests, questionnaires and interviews.
- Analyses adjusted for prior knowledge and time on task. Findings showed different effects on conceptual understanding and transfer, informing recommendations to align interaction design with learning objectives.

Teaching and Educational Technology

Teaching Assistant, Newcastle University *2026 – Present*

- **2026:** Human–Artificial Intelligence (AI) Interaction & Futures (CSC8611); delivered practical sessions and several lectures, and marked coursework. Completed ILTHE.
- **2027:** Programming Portfolio 1 (CSC1034); Computer Systems Design and Architectures (CSC1032); Biomedical Data Analytics and AI (CSC3432).

Teaching Support, Jiangxi University of Finance and Economics *Feb – Jun 2020*

Assisted with Python and Data Structures courses; demonstrated code and helped students debug programming exercises.

Product Manager Intern, NetEase Youdao — Super Dictionary *Sep – Dec 2020*

Conducted user research for an AI-assisted learning device (3,000 survey responses, 20 interviews and a 105-participant quantitative study); translated findings into learner personas, digital-library and learning-incentive features.

Methods, Skills and Service

Qualitative and design: reflexive thematic analysis (RTA), interviews, focus groups, cultural probes and participatory design.
Quantitative: controlled and crossover studies, questionnaires, statistical testing, causal inference, heterogeneous treatment effects, mediation and moderation analysis.
Learning frameworks: TPACK (Technological Pedagogical Content Knowledge); Bloom's taxonomy.
Implementation: Python, RAG, LLM integration and evaluation, interactive web prototypes, Unity/VR.
Peer review: CHI 2027, NIME 2026, EDM 2025, IUI 2025.
Languages: Chinese (native); English (TOEFL 109).

Publications

1. Yidan Zhu, **Xin Shu**, Lei Shi. From Observer to Creator: Unpacking the Tradeoff between Conceptual Understanding and Transfer in VR Cultural Heritage Learning. *IJHCI, 2026*.
2. **Xin Shu** et al. FretFlow: Adaptive Haptics for Rhythm and Articulation in Guitar Learning. *CHI 2026*.
3. Yunfei Wang et al. (including **Xin Shu**). Quiz Traveller: A Crowdsourced Educational Game on Chat-Based Interactive Live-Streaming Platforms. *GamiFIN 2026. Best Paper Award*.
4. Xixiang Nie et al. (including **Xin Shu**). ShadowAI: Fostering Children's AI Literacy and Critical Understanding of AI Reasoning. *C&C 2026*. Poster.
5. Xixiang Nie, **Xin Shu**, Matthew Wood. Casting the Shadow for Understanding "Reasoning Ghosts". *C&C 2026*. Technical demonstration.
6. Xixiang Nie, **Xin Shu**. Somaditation: A Multisensory VR Meditation Experience with a Wearable Glove. *TEI 2026*. Student Design Challenge.
7. **Xin Shu** et al. FretMate: ChatGPT-Powered Adaptive Guitar Learning Assistant. *IUI 2025*.
8. Xinrui Lin et al. (including **Xin Shu**). Seeking Inspiration through Human–LLM Interaction. *CHI 2025*.
9. Haowei Xu, **Xin Shu**. GlobalUP: Leveraging Social Media to Provide Social Support for Chinese Cultural Groups in Unfamiliar Environments. *HCII 2025*.
10. Mengdi Chu et al. (including **Xin Shu**). Work with AI and Work for AI: Autonomous Vehicle Safety Drivers' Lived Experiences. *CHI 2023*.
11. Jiangtao Gong et al. (including **Xin Shu**). Contactless Physiological Sensing in Smart Home Environments. *Packaging Engineering, 2022*, 43(16), 10–26.